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Wire harness mounting structure

Abstract

A wire harness mounting structure is provided for a vehicle including a drive unit and a battery unit configured to supply electric power to the drive unit via a wire harness. The drive unit includes a drive motor configured to drive a drive wheel, and an inverter device configured to convert the electric power from the battery unit and supply the converted electric power to the drive motor. One end of the wire harness is coupled to the inverter device from a first direction in a vehicle width direction via an inverter-side connector at a side surface portion of the inverter device, and the other end of the wire harness is coupled to the battery unit from a second direction opposite to the first direction via a battery-side connector at a rear end portion of the battery unit.

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Background/Summary

TECHNICAL FIELD
(1) The present invention relates to a wire harness mounting structure.
BACKGROUND ART
(2) An electric vehicle runs by converting electric power supplied from a battery into electric power suitable for driving a drive motor by an inverter device and driving the drive motor.

(3) JP5589772B discloses an electric vehicle in which a power unit (a drive motor) and an inverter device are disposed in a motor room, and the inverter device and a main battery disposed underneath a floor of the vehicle are connected via a power cable (a wire harness) routed in a front-rear direction of the vehicle.

SUMMARY OF INVENTION

Problem to be Solved by the Invention

(4) In recent years, a battery tends to become larger in order to extend a cruising distance of a vehicle, and a battery unit is disposed over an entire underfloor portion between front wheels and rear wheels of the vehicle. In such a configuration, since a layout space is small near the rear wheels in particular, it is necessary to devise the arrangement of a drive motor, an inverter device, and a harness. Further, in order to cope with swinging of the drive motor, it is also required to provide a margin in a harness length.

(5) In the above patent document, a wire harness connecting a main battery to an inverter device is routed in a front-rear direction of a vehicle. However, it is difficult to form a margin space in the front-rear direction of the vehicle due to factors such as an increase in size of the battery. Therefore, there is a problem that it is not easy to provide a margin in a harness length in the front-rear direction of the vehicle, and it is difficult to cope with the swinging of the drive unit.

(6) The present invention has been made in view of such these problems. An object of the present invention is to provide a wire harness mounting structure capable of providing a margin in a length of a wire harness particularly on a vehicle rear side which has a small space.

Means for Solving the Problem

(7) According to an embodiment of the present invention, provided is a wire harness mounting structure for a vehicle, the vehicle including a drive unit configured to drive the vehicle and a battery unit configured to supply electric power to the drive unit via a wire harness. The battery unit is disposed on a floor between a front wheel and a rear wheel of the vehicle. The drive unit includes a drive motor disposed on a rear side of the vehicle relative to the battery unit and configured to drive a drive wheel, and an inverter device configured to convert the electric power from the battery unit and supply the converted electric power to the drive motor. The wire harness is configured such that one end is coupled to the inverter device from a vehicle width direction at a side surface portion of the inverter device, and the other end is coupled to the battery unit from the vehicle width direction at a rear end portion of the battery unit.

Effect of the Invention

(8) According to the present invention, the wire harness is coupled to the inverter device and the battery unit in the vehicle width direction. Accordingly, even when the drive unit and the battery unit are disposed close to each other, the wire harness can be routed in a lateral direction in a gap between the drive unit and the battery unit. Further, since a harness length of the wire harness can be provided with a margin in the lateral direction, a stress of the wire harness when the drive unit swings can be relaxed.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1** is an explanatory diagram of a drive unit according to an embodiment of the present invention.

(2) FIG. **2** is an explanatory diagram of an electric vehicle on which the drive unit is mounted.

(3) FIG. **3** is an explanatory diagram of the drive unit.

(4) FIG. **4** is an explanatory diagram of the drive unit.

(5) FIG. **5** is a cross-sectional view of the electric vehicle.

DESCRIPTION OF EMBODIMENTS

(6) Hereinafter, an embodiment of the present invention will be described with reference to the drawings or the like.
(7) FIG. **1** is an explanatory diagram of a rear drive unit **20** of the present embodiment. FIG. **2** is an explanatory diagram of an entire drive system of an electric vehicle **1** on which the rear drive unit **20** according to the embodiment of the present invention is mounted.
(8) As shown in FIG. **2**, the electric vehicle **1** (hereinafter, referred to as a vehicle **1**) includes the rear drive unit **20**, a battery unit **30**, and a front drive unit **40**. These units **20**, **30**, and **40** are mounted on a vehicle body **10** of the electric vehicle **1**.
(9) The rear drive unit **20** is disposed on the rear side of the electric vehicle **1**, rather than the battery unit **30**. A wire harness **35** for supplying DC power of the battery unit **30** is connected to the rear drive unit **20**.
(10) As shown in FIG. **1**, the rear drive unit **20** includes an inverter device **21**, a drive motor **22**, and a speed reducer **23**. A cooling water pipe **26** through which cooling water flows is connected to the rear drive unit **20**. The rear drive unit **20** drives the vehicle **1** by driving the drive motor **22** based on the electric power from the battery unit **30** to rotate an axle **23***a* and a rear wheel **15** (see FIG. **5**) which is a drive wheel.
(11) In the rear drive unit **20**, the drive motor **22** and the speed reducer **23** are disposed in a vehicle width direction such that rotation shafts thereof are parallel to each other. The inverter device **21** is disposed above the drive motor **22** and the speed reducer **23** so as to be inclined to a front side of the vehicle **1**.
(12) The inverter device **21** includes an inverter (not shown) that converts the electric power supplied from the battery unit **30** into electric power suitable for driving the drive motor **22** and supplies the electric power to the drive motor **22**. The inverter is housed in a housing **24** that is a metal or resin case. The inverter device **21** receives electric power from the battery unit **30** and drives the drive motor **22**. The inverter device **21** receives regenerative electric power from the drive motor **22** and charges the battery unit **30**.
(13) The drive motor **22** includes a rotating electric machine such as a permanent magnet embedded synchronous motor or a field winding synchronous motor.
(14) The speed reducer **23** includes a plurality of gears that reduce a speed of rotation of the drive motor **22** and transmit the rotation to the axle. The speed reducer **23** may be configured as a transmission capable of switching a plurality of gear positions stepwise or steplessly.
(15) As shown in FIG. **2**, the battery unit **30** is disposed on a floor between a front wheel and a rear wheel of the vehicle **1**. The battery unit **30** includes a plurality of battery modules therein. The wire harness **35** for supplying electric power is connected to the inverter device **21** and the battery unit **30**.
(16) The wire harness **35** includes a pair of positive and negative conducting wires obtained by applying an insulating coating to a conductor made of copper, aluminum, or the like. The wire harness **35** is covered with a protector **60** and a cushioning material **61**. The wire harness **35** includes terminals connected to the inverter device **21** and the battery unit **30** at both ends thereof, is routed in the vehicle width direction between the terminals, and is routed in a curved S-shape in a top view.
(17) As shown in FIG. **2**, the front side (a motor room) of the vehicle **1** is provided with the front drive unit **40** that drives a front wheel. Similarly to the rear drive unit **20**, the front drive unit **40** includes a front drive motor (not shown) and a front inverter device **41** that supplies electric power to the front drive motor. A front wire harness **36** for supplying electric power is connected between the front inverter device **41** of the front drive unit **40** and the battery unit **30**.
(18) The vehicle **1** of the present embodiment may not necessarily include the front drive unit **40**. The vehicle **1** may be configured to drive only the rear wheel **15** by the rear drive unit **20**.
(19) Next, the wire harness **35** routed between the rear drive unit **20** and the battery unit **30** will be described.

(20) When the vehicle **1** travels, the rear drive unit **20** swings relative to the battery unit **30** fixed to the vehicle body **10** due to vibration input from a road surface via a suspension or the like, vibration of the rear drive unit **20** itself, or the like. In order to prevent the wire harness connected therebetween from being deformed and damaged due to the stress caused by the swinging of the rear drive unit **20**, it is desirable to provide a margin length to allow the swinging.
(21) On the other hand, in the rear drive unit **20** disposed on the vehicle rear side, the battery unit **30** is disposed on a front side thereof, and a structure of the vehicle body **10** such as a luggage compartment is present on an upper side thereof, so that a height of the vehicle body **10** from the ground is low. Due to the structure, there is no sufficient margin in a space between the rear drive unit **20** and the battery unit **30**. Further, since the wire harness is configured with thick wires so that the voltage (for example, DC 400V) of the battery unit **30** flows therethrough, the degree of freedom in bending and elasticity are low. For this reason, in the related art, there is a problem that it is difficult to provide a wire harness with a margin length.
(22) In the present embodiment, in order to solve such a problem, the wire harness **35** is configured to have a margin length as described below.
(23) As shown in FIG. **1**, the wire harness **35** is configured such that one end is connected to an inverter-side connector **211** provided on a side surface of the inverter device **21**, and the other end is connected to a battery-side connector **311** provided in a connecting portion **31** arranged to protrude from a rear end side of the battery unit **30**. The one end of the wire harness **35** is configured as a one-end-side connection terminal P**1**, and the other end thereof is configured as the other-end-side connection terminal P**2**. The housing **24** of the inverter device **21** includes an opening **241** to which the one-end-side connection terminal P**1** described in detail with reference to FIG. **3** is fixed, and a cover **242** that closes the opening **241**.
(24) The inverter-side connector **211** is provided on a left side of the inverter device **21** in a front-rear direction of the vehicle, and is connected to the one-end-side connection terminal P**1** of the wire harness **35**.
(25) The battery-side connector **311** is provided on a right side of the connecting portion **31** protruding from the rear end side of the battery unit **30** in the front-rear direction of the vehicle. The other-end-side connection terminal P**2** of the wire harness **35** is connected to the battery-side connector **311**.
(26) After extending in the vehicle width direction from the inverter-side connector **211** on a left side surface of the inverter device **21**, the wire harness **35** is bent by 180° toward the right side in the vehicle width direction. The wire harness **35** is routed from the left side to the right side in the vehicle width direction between the inverter device **21** and the battery unit **30**, and is bent by 180° toward the left side in the vehicle width direction near a right side surface of the inverter device **21**. The wire harness **35** extends to the battery-side connector **311** disposed on a right side surface of the connecting portion **31** of the battery unit **30**.
(27) As described above, the wire harness **35** is configured such that one end is coupled to the inverter-side connector **211** provided on the left side surface of the inverter device **21** from the left side to the right side (a first direction) in the vehicle width direction, and the other end is coupled to the battery-side connector **311** provided on the right side surface of the connecting portion **31** of the battery unit **30** from the right side to the left side (a second direction) in the vehicle width direction. Accordingly, the wire harness **35** is routed in the vehicle width direction and is formed in an S-shape (a Z-shape) bent at about 180° in a top view.
(28) Also in the front inverter device **41** of the front drive unit **40**, one end of the wire harness **36** is coupled to a connector provided on a left side surface of the front inverter device **41** from the left side to the right side (the second direction) in the vehicle width direction.
(29) In addition, the wire harness **35** is covered with a U-shaped hard resin protector **60** so as to cover an outer periphery of the wire harness **35** near the inverter-side connector **211**. The protector **60** is supported by an L-shaped bracket **65** formed on the housing **24** of the inverter device **21** so as

to maintain a predetermined distance from the inverter device **21**.
(30) The bracket **65** includes a housing fixing portion fixed to a surface of the housing **24** by bolting, and a protector fixing portion extending toward the protector **60** and fixed to a back surface of the protector **60** by bolting. The housing fixing portion and the protector fixing portion are formed in a crank shape in the front-rear direction of the vehicle.
(31) The wire harness **35** is covered with a cushioning material **61** made of soft resin near the battery-side connector **311**.
(32) The wire harness **35** is restricted from swinging by the protector **60** and the bracket **65** near the inverter device **21** while maintaining a predetermined distance from the inverter device **21**. Accordingly, the stress caused by the vibration of the wire harness **35** more than necessary is suppressed from acting, and the durability of the wire harness **35** is improved. On the other hand, on the battery-side connector **311** side of the wire harness **35**, the soft cushioning material **61** allows swinging, and suppresses the wire harness **35** from being buffered with other parts.
(33) With such a configuration, the wire harness **35** can have a margin length in the vehicle width direction while the rear drive unit **20** is disposed close to the battery unit **30**. Accordingly, since the wire harness **35** is configured to be swingable, relative swinging between the rear drive unit **20** and the vehicle body **10** to which the battery unit **30** is fixed is allowed, and the stress applied to the wire harness **35** can be relaxed.
(34) Next, a configuration of the one-end-side connection terminal P**1** of the wire harness **35** will be described.
(35) FIGS. **3** and **4** are explanatory diagrams of the inverter device **21** of the present embodiment, and are diagrams illustrating the inverter-side connector **211**. FIG. **3** is a perspective view of the inverter device **21**, and FIG. **4** is a front view of the inverter device **21** when viewed from the front side of the vehicle.
(36) FIGS. **3** and **4** both show a state in which the protector **60** of the wire harness **35** and the cover **242** of the inverter-side connector **211** are removed. The speed reducer **23** is omitted.
(37) An insertion hole **215** into which the one-end-side connection terminal P**1** is inserted is formed in a side surface of the inverter device **21**. On a front surface side of the inverter device **21**, an opening **241** for bolting in a state in which the one-end-side connection terminal P**1** is inserted is formed. A terminal (not shown) is provided inside the opening **241**, and the one-end-side connection terminal P**1** is fixed to the terminal.
(38) A plate-shaped positive electrode terminal BP**1** connected to a conducting wire on a positive electrode side and a plate-shaped negative electrode terminal BN**1** fixed to a conducting wire on a negative electrode side are provided at a tip of the one-end-side connection terminal P**1** of the wire harness **35**. A bolt hole for fixing a bolt is formed in each terminal. The one-end-side connection terminal P**1** of the wire harness **35** is provided with a boot **214** fixed to the insertion hole **215**. As described above, in the inverter-side connector **211**, the positive electrode terminal BP**1** and the negative electrode terminal BN**1** of the one-end-side connection terminal P**1** are connected to terminals in the inverter device **21** through the opening **241**.
(39) The inverter-side connector **211** and the one-end-side connection terminal P**1** configured as described above are assembled as follows. First, the positive electrode terminal BP**1** and the negative electrode terminal BN**1** constituting the one-end-side connection terminal P**1** of the wire harness **35** are inserted from the insertion hole **215** opened in the side surface of the housing **24**. The positive electrode terminal BP**1** and the negative electrode terminal BN**1** are fixed to the terminals on the inverter device **21** side through the opening **241** by bolting. After the positive electrode terminal BP**1** and the negative electrode terminal BN**1** are fixed, the opening **241** is closed by the cover **242**. The boot **214** sheathing the one end of the wire harness **35** is fixed to the insertion hole **215** by bolting.
(40) In this way, the one-end-side connection terminal P**1** is fixed to the inverter-side connector **211**.

(41) As shown in FIG. **3**, the other end side of the wire harness **35** includes the other-end-side connection terminal P**2** which is a one-touch connector that can be inserted into and removed from the connecting portion **31** of the battery unit **30**. The other-end-side connection terminal P**2** is locked by being inserted into the battery-side connector **311**.
(42) With such a configuration, when the rear drive unit **20** is detached from the vehicle body **10** at the time of maintenance of the vehicle **1**, firstly, an operator detaches the other-end-side connection terminal P**2** from the battery-side connector **311** of the battery unit **30**. The other-end-side connection terminal P**2** is a one-touch connector, and thus can be easily attached and detached in a predetermined procedure. The rear drive unit **20** is detached from the vehicle body **10** in a state in which the wire harness **35** is connected thereto.
(43) Next, a relationship between the vehicle body **10** and the rear drive unit **20** will be described.
(44) FIG. **5** is a longitudinal cross-sectional view of the vehicle **1** in the front-rear direction centering on the rear drive unit **20** of the present embodiment.
(45) The rear drive unit **20** is disposed in an underfloor portion on the rear side of the vehicle body **10** via a suspension member **102** fixed to the vehicle body **10**. A cross member **101** that crosses the vehicle body **10** in the width direction is disposed in the vehicle body **10**. The rear drive unit **20** is disposed adjacent to the cross member **101** and behind the cross member **101**. The battery unit **30** is fixed to the underfloor portion of the vehicle body **10** on a front side of the cross member **101**.
(46) The above-described wire harness **35** is routed between the rear drive unit **20** and the battery unit **30**. The wire harness **35** is covered with the protector **60** near the inverter device **21**. The protector **60** is fixed to an outside of the housing **24** of the inverter device **21** by the bracket **65**.
(47) Here, a case where the vehicle **1** collides is considered. In particular, in a case where the vehicle **1** collides in a traveling direction, the vehicle body **10** and the suspension member **102** are deformed near the rear drive unit **20**, and the rear drive unit **20** moves toward the front side. Due to the deformation, the inverter device **21** disposed on the front side of the rear drive unit **20** may move to the cross member **101**.
(48) In this case, since the rear drive unit **20** and the cross member **101** approach each other, the wire harness **35** may be sandwiched between the housing **24** of the inverter device **21** and the cross member **101**.
(49) Therefore, in the present embodiment, a fixing position of the bracket **65** that fixes the protector **60**, that is, the housing fixing portion fixed to the housing **24** and the protector fixing portion fixed to the protector **60** is lower than a height of the cross member **101** from the ground (a bottom surface of the cross member **101** indicated by a dash-dotted line in FIG. **5**). As described above, by fixing the bracket **65** at a position lower than the cross member **101**, a position of the wire harness **35** is restricted by the protector **60** when the vehicle **1** collides. Accordingly, since the wire harness **35** does not move upward, the wire harness **35** is suppressed from being sandwiched between the housing **24** and the cross member **101** at the time of a collision.
(50) As described above, the vehicle **1** according to the embodiment of the present invention includes a drive unit (the rear drive unit **20**) and the battery unit **30** that supplies electric power to the rear drive unit **20** via the wire harness **35**. The battery unit **30** is disposed on the floor between the front wheel and the rear wheel of the vehicle **1**, and the rear drive unit **20** is disposed on the rear side of the vehicle **1** rather than the battery unit **30**. The rear drive unit **20** includes the drive motor **22** that drives the rear wheel **15** which is a drive wheel, and the inverter device **21** that converts the electric power from the battery unit **30** and supplies the converted electric power to the drive motor **22**. The wire harness **35** is configured such that one end is coupled to the inverter device **21** from the vehicle width direction at a side surface portion of the inverter device **21**, and the other end is coupled to the battery unit **30** from the vehicle width direction at a rear end portion of the battery unit **30**.
(51) With these configurations, the wire harness **35** is coupled to the inverter device **21** and the battery unit **30** in the vehicle width direction. Accordingly, even when the inverter device **21** and

the battery unit **30** are disposed close to each other, the wire harness **35** can be routed in the lateral direction of the vehicle **1**. Further, since a harness length of the wire harness **35** can have a margin in the lateral direction between the inverter device **21** and the battery unit **30**, the stress applied to the wire harness **35** can be relaxed when the rear drive unit **20** swings.
(52) In the present embodiment, the one end of the wire harness **35** is coupled to the inverter device **21** via the inverter-side connector **211** from the left side to the right side (the first direction) in the vehicle width direction, and the other end thereof is coupled to the battery unit **30** via the battery-side connector **311** from the right side to the left side (the second direction) in the vehicle width direction, which is opposite to the first direction.
(53) With these configurations, since the wire harness **35** is bent and routed in an S-shape between the first direction and the second direction, the wire harness **35** can be routed in the lateral direction in the gap between the inverter device **21** and the battery unit **30**, and the harness length of the wire harness **35** can be provided with a margin.
(54) In the present embodiment, the inverter-side connector **211** fixes the wire harness **35** to the inverter device **21** by bolting, and the battery-side connector **311** includes a detachable one-touch connector.
(55) With these configurations, the rear drive unit **20** can be detached from the vehicle body **10** together with the wire harness **35** by detaching the battery-side connector **311** that can be easily attached and detached at the time of maintenance or the like.
(56) In the present embodiment, a front side of the vehicle **1** is provided with the front drive unit **40** including the front inverter device **41** and the front wire harness **36** that supplies the electric power from the battery unit **30** to the front inverter device **41**. The front wire harness **36** is configured such that one end is coupled to the battery unit **30**, and the other end is coupled to the front inverter device **41** from the left side to the right side (the first direction) in the vehicle width direction.
(57) In this way, since the inverter device **21** disposed on a rear wheel side and the front inverter device **41** disposed on a front wheel side are connected to the wire harnesses **35** and **36** from the same first direction, the inverter device **21** disposed on the rear side and the front inverter device **41** can have the same connector and internal configuration. Accordingly, there is no need to use different parts for the inverter device **21** and the front inverter device **41**, and manufacturing costs can be suppressed.
(58) In the present embodiment, the wire harness **35** is covered with the protector **60** to restrict swinging near the one end, that is, near the inverter-side connector **211** of the inverter device **21**.
(59) With this configuration, the wire harness **35** is restricted from swinging more than necessary near the inverter-side connector **211** of the inverter device **21**, and the durability of the wire harness **35** can be improved. On the battery-side connector **311** side of the wire harness **35**, the soft cushioning material **61** allows swinging, and suppresses the wire harness **35** from being buffered with other parts.
(60) In the present embodiment, the rear drive unit **20** is provided adjacent to the cross member **101** that crosses underneath the floor of the vehicle **1** in the vehicle width direction and behind the cross member **101**, the protector **60** is fixed to the housing **24** of the inverter device **21** by the bracket **65**, and a fixing position between the bracket **65** and the housing **24** is located between the cross member **101** and the inverter device **21** and below the cross member **101**.
(61) With these configurations, in a case where the inverter device **21** and the cross member **101** approach each other due to deformation of the vehicle body **10** at the time of a vehicle collision, the wire harness **35** can be prevented from being sandwiched between the inverter device **21** and the cross member **101** by the fixing position of the protector **60** and the bracket **65**.
(62) Although the embodiment and modifications of the present invention (the wire harness attaching (connecting) structure) have been described above, the above embodiment and modifications are merely a part of application examples of the present invention, and do not mean

that the technical scope of the present invention is limited to the specific configurations of the above embodiment.

Claims

1. A system comprising: a wire harness; a drive unit configured to drive a vehicle; and a battery unit configured to supply electric power to the drive unit, wherein the battery unit is located on a front side of the drive unit, wherein: the drive unit comprises a drive motor configured to drive a drive wheel, and an inverter device located on a front side of the drive motor and configured to convert the electric power from the battery unit and supply the converted electric power to the drive motor, and the wire harness is formed in a continuous S-shape and configured such that one end of the wire harness is coupled to the inverter device from a first direction in a vehicle width direction via an inverter-side connector at a side surface portion of the inverter device, and the other end of the wire harness is coupled to the battery unit from a second direction opposite to the first direction via a battery-side connector at a rear end portion of the battery unit.
2. The system according to claim 1, wherein: the inverter-side connector is configured to fix the wire harness to the inverter device by bolting, and the battery-side connector comprises a detachable one-touch connector.
3. The system according to claim 2, wherein: a front side of the vehicle comprises a front drive unit comprising a front inverter device, and a front wire harness configured to supply the electric power from the battery unit to the front inverter device, and wherein: the front wire harness is configured such that one end of the front wire harness is coupled to the battery unit, and the other end of the front wire harness is coupled to the front inverter device from the first direction.
4. The system according to any claim 1, further comprising: a protector that covers an end portion the wire harness, the protector configured to restrict swinging of the wire harness.
5. A vehicle comprising: a cross member that crosses underneath a floor of the vehicle in a vehicle width direction; and the system according to claim 4, wherein: the drive unit is located adjacent to and behind the cross member, the protector is fixed to a housing of the inverter device by a bracket, and a fixing position between the bracket and the housing is located between the cross member and the inverter device and below the cross member.